

Single-Photon Molecular Recoil as a Direct Test of the SAT/H(s)H Kinetic Coupling

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Abstract

A recent experiment detected absorption of a single mid-infrared photon by a single $^{40}\text{CaOH}^+$ molecular ion through the recoil imparted to a co-trapped two-ion crystal. This provides an unusually clean system in which a known quantum of momentum is converted into measurable motion. Here the SAT/H(s)H master Lagrangian ansatz is reduced to the recoil degree of freedom and compared with the standard momentum-transfer result. The ansatz reproduces the conventional recoil exactly when the recoil coordinate is orthogonal to the distinguished time-direction coupling or when that coupling vanishes. If the finite worldtube geometry permits a nonzero projection of the laboratory recoil direction onto that coupling, the same kinetic term yields a simple orientation-dependent modification of the inertial response. The result is a null-and-deviation test: conventional recoil is the baseline prediction, while any SAT/H(s)H-specific effect must appear as a fixed angular modulation whose amplitude is controlled by a previously determined coupling, not fitted to the recoil data.

1 Experimental opportunity

Wu *et al.* recently demonstrated non-destructive infrared absorption spectroscopy on a single $^{40}\text{CaOH}^+$ molecular ion co-trapped with $^{40}\text{Ca}^+$ [1]. The absorption of one photon is detected through the momentum transferred to the molecule. Coulomb coupling transfers the recoil into the shared in-phase motional mode of the two-ion crystal, and a non-classical motional cat state amplifies the resulting displacement so that it can be read out through the atomic ion.

For a photon with wavevector component k_z along the measured motional axis, standard momentum conservation gives

$$J_\gamma = \Delta p = \hbar k_z. \quad (1)$$

The experiment describes the corresponding motional displacement with a Lamb–Dicke parameter

$$\eta_m = \sqrt{\frac{\hbar}{2M_m\omega_z}} k_{z,m} e_{z,m}, \quad (2)$$

where M_m is the molecular mass, ω_z the in-phase mode frequency, and $e_{z,m}$ the molecular participation factor [1].

This is a useful test system because the input impulse is fixed by the absorbed photon. A candidate dynamical framework is therefore not asked to invent the recoil; it is asked to specify how the system responds to a known impulse.

2 SAT/H(s)H kinetic sector

The SAT/H(s)H working specification writes the master Lagrangian schematically as [3]

$$L = \sum_{i,o} \left[\frac{1}{2} \mu_i \left| \frac{d\lambda}{dH} \right|^2 + \sum_j \kappa F_{\text{braid}} + \alpha \left(\frac{d\lambda}{dH} \cdot T \right)^2 + V_{\text{geom}}(\lambda) \right]. \quad (3)$$

Here $\lambda(H)$ denotes a worldline/worldtube coordinate, T is the distinguished time-direction object appearing in the coupling term, and α controls the quadratic projection onto T . The present calculation uses only the kinetic block. It does not require a choice of braid functional or geometric potential because the photon absorption is treated as an external impulsive interaction over a time short compared with the motional evolution.

For a local recoil coordinate q along a unit laboratory direction $\mathbf{n}$, write

$$\lambda' = q' \mathbf{n}, \quad \chi \equiv \mathbf{n} \cdot T, \quad (4)$$

where a prime denotes differentiation with respect to the evolution parameter. Up to a constant affine reparameterization of H , which rescales both quadratic coefficients together, the reduced kinetic Lagrangian is

$$L_{\text{kin}} = \frac{1}{2} M \dot{q}^2 + \alpha_T \chi^2 \dot{q}^2. \quad (5)$$

The canonical momentum is therefore

$$p_q = \frac{\partial L}{\partial \dot{q}} = (M + 2\alpha_T \chi^2) \dot{q}. \quad (6)$$

It is convenient to define the dimensionless coupling

$$\zeta \equiv \frac{2\alpha_T}{M}. \quad (7)$$

Then

$$p_q = M (1 + \zeta \chi^2) \dot{q}. \quad (8)$$

3 Single-photon recoil prediction

Across an impulsive absorption event, the known photon impulse fixes the jump in canonical momentum,

$$\Delta p_q = \hbar k_z. \quad (9)$$

The SAT/H(s)H reduced response is therefore

$$\boxed{\Delta v_{\text{HsH}} = \frac{\hbar k_z}{M (1 + \zeta \chi^2)}} \quad (10)$$

while the standard result is

$$\Delta v_{\text{std}} = \frac{\hbar k_z}{M}. \quad (11)$$

Hence the directly testable ratio is

$$\boxed{\frac{\Delta v_{\text{HsH}}}{\Delta v_{\text{std}}} = \frac{1}{1 + \zeta \chi^2}} \quad (12)$$

and, for $|\zeta\chi^2| \ll 1$,

$$\frac{\Delta v_{\text{HsH}} - \Delta v_{\text{std}}}{\Delta v_{\text{std}}} \simeq -\zeta\chi^2. \quad (13)$$

This gives two sharply separated predictions.

Null sector. If the laboratory recoil direction is orthogonal to T in the relevant reduced geometry, $\chi = 0$, then

$$\Delta v_{\text{HsH}} = \Delta v_{\text{std}}. \quad (14)$$

The same is true if $\alpha_T = 0$. Thus the ansatz is not expected to produce an arbitrary recoil anomaly.

Deviation sector. If finite worldtube geometry produces a nonzero effective projection χ , the ansatz predicts an even angular modulation proportional to χ^2 . For a simple directional projection $\chi = \cos\theta$,

$$\Delta v(\theta) = \frac{\hbar k_z / M}{1 + \zeta \cos^2 \theta}. \quad (15)$$

The modulation has period π and is invariant under $\mathbf{n} \rightarrow -\mathbf{n}$. Its amplitude is not a free fit parameter in a valid test: ζ must be determined independently from the SAT/H(s)H construction or from a separate calibration system before the recoil data are examined.

4 Numerical baseline for $^{40}\text{CaOH}^+$

The reported spectroscopy includes measurements near 3703 cm^{-1} and targets an O–H stretch predicted near 3783 cm^{-1} [1]. Taking 3703 cm^{-1} as a representative absorbed photon gives

$$\tilde{\nu} = 3.703 \times 10^5\text{ m}^{-1}, \quad (16)$$

so

$$p_\gamma = \hbar\tilde{\nu} = 2.45363 \times 10^{-28}\text{ kg m s}^{-1}. \quad (17)$$

Using $M \simeq 57u$ for $^{40}\text{CaOH}^+$ and the 2022 CODATA atomic-mass constant [2],

$$M \simeq 9.46507 \times 10^{-26}\text{ kg}, \quad (18)$$

which gives the free-ion standard recoil scale

$$\Delta v_{\text{std}} \simeq 2.592 \times 10^{-3}\text{ m s}^{-1}. \quad (19)$$

This number is only a convenient scale. The experiment reads the recoil through the normal modes of the two-ion crystal, so a quantitative experimental comparison should use the measured mode frequency and participation factors rather than a free-ion velocity.

Equation (12) is the cleaner prediction because the standard trap dynamics can divide out. For illustration only,

ζ	maximal fractional shift ($\chi^2 = 1$)	recoil scale
10^{-4}	$\simeq -10^{-4}$	$\sim 0.26\text{ }\mu\text{m s}^{-1}$
10^{-3}	$\simeq -10^{-3}$	$\sim 2.6\text{ }\mu\text{m s}^{-1}$
10^{-2}	$\simeq -9.9 \times 10^{-3}$	$\sim 26\text{ }\mu\text{m s}^{-1}$

These rows are sensitivity examples, not SAT/H(s)H parameter estimates.

5 Experimental test

A decisive test would use the same molecular transition and the same trap protocol while varying only the orientation entering χ . The analysis should be preregistered in the following order:

1. Determine ζ and the geometrical definition of χ independently of the recoil dataset.
2. Predict the complete angular response from Eq. (15), including null orientations.
3. Hold laser frequency, pulse sequence, trap mode, ion species, and analysis procedure fixed as far as experimentally possible.
4. Compare the measured recoil-sensitive phase or inferred displacement with the standard model and the frozen SAT/H(s)H curve.
5. Count both agreement and disagreement. No post-hoc change in ζ , angular map, or projection rule is admissible.

A useful control is reversal of the photon propagation direction. The quadratic coupling in Eq. (15) predicts the same magnitude for θ and $\theta + \pi$ after ordinary sign reversal of the recoil itself. An observed odd-parity anisotropy would therefore not be explained by this kinetic term alone.

6 Interpretation and scope

This calculation does not claim that the reported CaOH^+ experiment has already observed a deviation from standard recoil. It has not. The published experiment is consistent with ordinary momentum transfer and uses that recoil as the spectroscopy signal [1].

The point is narrower and falsifiable. When the SAT/H(s)H kinetic ansatz is instantiated on this system, it contains a conventional sector and a possible residual sector. The conventional sector gives $\Delta p = \hbar k$ and the ordinary recoil response. A departure arises only if the time-coupling term has a nonzero effective projection onto the measured recoil degree of freedom. In that case the ansatz predicts a specific even angular modulation, Eq. (15).

The strongest version of the test is therefore not “does SAT/H(s)H explain photon recoil?” Standard physics already does. It is:

Does the independently fixed SAT/H(s)H kinetic coupling predict a reproducible orientation-dependent residual in single-photon recoil after the conventional trap dynamics are accounted for?

A null result constrains $|\zeta\chi^2|$ directly. A detected modulation with the wrong symmetry falsifies this reduced prediction. A detected modulation with the predicted symmetry would still require systematic checks and independent replication before being interpreted as evidence for the model.

7 Conclusion

Single-photon molecular recoil provides a compact system-level test of the SAT/H(s)H master Lagrangian ansatz. The ansatz reproduces standard recoil in its transverse/null sector. If its finite worldtube

geometry gives the laboratory recoil coordinate a nonzero projection onto the T coupling, the leading response becomes

$$\Delta v / \Delta v_{\text{std}} = (1 + \zeta \chi^2)^{-1}.$$

The experimental task is therefore unusually clean: determine the model geometry and coupling first, freeze them, and then look for the predicted residual. The absence of such a residual is informative; the presence of an arbitrary residual is not sufficient. The value of the test lies in the fixed relation between geometry and recoil response.

References

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